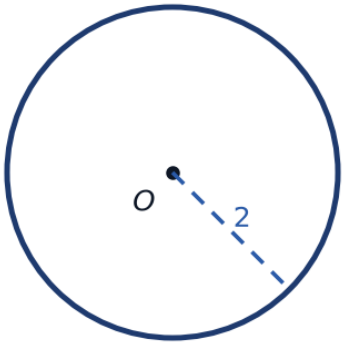
Polar Curves and Derivatives

ClipSAT Printable Practice Suite (clipsat.org) • Topic 4.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

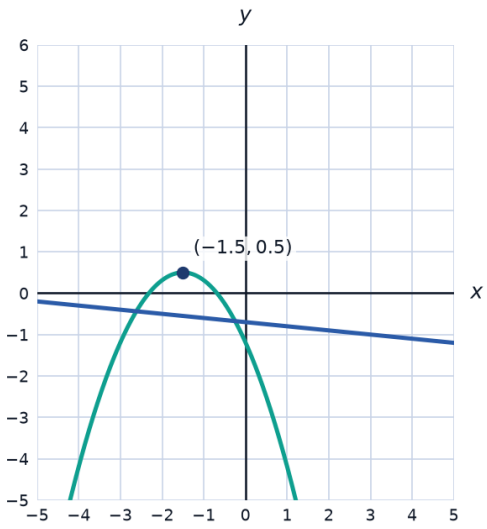
1 [Foundational] • Target: Converting between polar and rectangular coordinates

Convert the polar equation $r = 4\cos\theta$ to rectangular form, and identify the curve.



2 [Foundational] • Target: Identifying and sketching common polar curves

Identify the type of curve for $r = 2 + 2\sin\theta$, and state its key feature.



3 [Foundational] • Target: Identifying and sketching common polar curves

Identify the type of curve for $r = \cos(3\theta)$.

4 [Foundational] • Target: The slope of a polar curve

State the formula for $\frac{dy}{dx}$ of a polar curve $r = f(\theta)$, using $x = r\cos\theta$ and $y = r\sin\theta$.

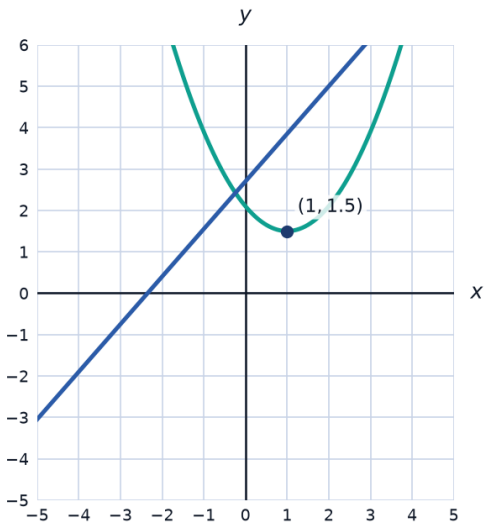
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5 [Foundational] • Target: Finding horizontal and vertical tangents

For the circle $r = 4\cos\theta$, find the value(s) of θ in $[0, \pi]$ where the tangent line is horizontal (where $y = r\sin\theta$ has a critical point).

6 [Foundational] • Target: Symmetry of polar curves

Explain how to test whether a polar curve $r = f(\theta)$ is symmetric about the polar axis (the x-axis).



7 [Foundational] • Target: Symmetry of polar curves

Test whether $r = 3\sin\theta$ is symmetric about the y-axis (replace θ with $\pi - \theta$).

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: More polar-to-rectangular conversions

Convert the rectangular point $(3, 3\sqrt{3})$ to polar coordinates ($r > 0$, $0 \le \theta < 2\pi$).

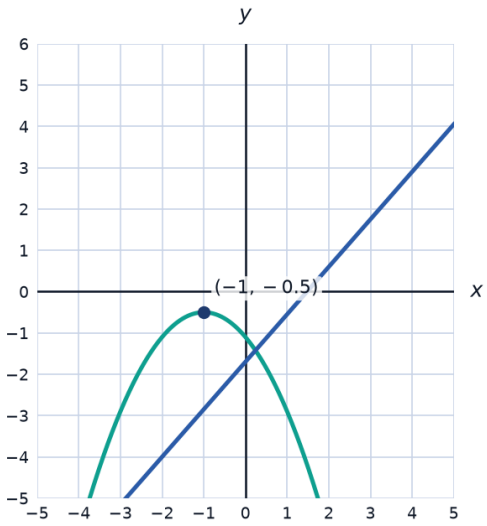
9 [Intermediate] • Target: More polar-to-rectangular conversions

Convert the rectangular equation $y = x$ to polar form.

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10 [Intermediate] • Target: More slope-of-a-polar-curve practice

Find $\frac{dy}{dx}$ for the cardioid $r = 1 + \sin\theta$ at $\theta = 0$.



11 [Intermediate] • Target: More slope-of-a-polar-curve practice

Find $\frac{dy}{dx}$ for the circle $r = 5$ (constant) at any angle θ, and interpret the result using the fact that a circle centered at the origin has radial symmetry.

12 [Intermediate] • Target: Tangent lines at the pole

Find the values of θ where the cardioid $r = 1 - \cos\theta$ passes through the pole ($r = 0$), and explain why these give the tangent line directions at the pole.

13 [Intermediate] • Target: Tangent lines at the pole

Find the values of $\theta \in [0, 2\pi)$ where the rose $r = \cos(2\theta)$ passes through the pole.

14 [Intermediate] • Target: Classifying polar curves by equation type

Classify each polar equation by curve type (circle, cardioid, limaçon with inner loop, rose, or line):

a $r = 5$

b $r = 3 + 3\cos\theta$

c $r = 1 + 3\sin\theta$

d $r = 4\sin(5\theta)$

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

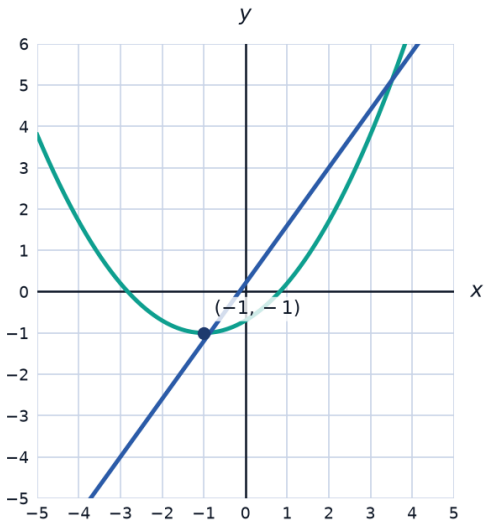
15 [Challenging] • Target: Number of petals in rose curves • Cross-Lesson: 6.02 Taylor and Maclaurin Series

State the rule for the number of petals of $r = a\cos(n\theta)$ or $r = a\sin(n\theta)$ depending on whether n is odd or even.

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16 [Challenging] • Target: Number of petals in rose curves • Cross-Lesson: 1.01 Integration by Parts

How many petals does $r = 2\sin(4\theta)$ have? How many does $r = 5\cos(7\theta)$ have?



17 [Challenging] • Target: Graphing polar curves point-by-point • Cross-Lesson: 1.02 Partial Fractions and Improper Integrals

Complete a short table of (θ, r) values for $r = 2 + 2\cos\theta$ at $\theta = 0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}$, and use it to describe the cardioid's shape.

18 [Challenging] • Target: Concavity considerations for polar curves • Cross-Lesson: 2.01 Euler's Method and Logistic Growth

Explain, in terms of $\frac{dy}{dx}$, what a 'vertical tangent' means for a polar curve, and why it corresponds to $\frac{dx}{d\theta} = 0$ (with $\frac{dy}{d\theta} \neq 0$).

19 [Challenging] • Target: More slope-of-a-polar-curve practice • Cross-Lesson: 3.01 Parametric Equations

Level 2 · MediumFind the angle(s) $\theta \in [0, \pi]$ at which the polar curve $r = 3\cos\theta$ passes through the origin, and determine the equation of the tangent line at that point.

20 [Challenging] • Target: Tangent lines at the pole • Cross-Lesson: 3.02 Vector-Valued Functions

Level 2 · MediumFind the slope of the tangent line $\frac{dy}{dx}$ to the polar cardioid $r = 1 - \cos\theta$ at $\theta = \frac{\pi}{2}$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Converting between polar and rectangular coordinates

How many petals does the rose $r = \cos(2\theta)$ have?

a. 4

b. 6

c. 7

d. 2

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22 [Foundational] • Target: Identifying and sketching common polar curves

Find the smallest positive θ where the rose $r = 3\sin(4\theta)$ passes through the pole.

a. $\frac{\pi}{4}$

b. $\frac{\pi}{16}$

c. $\frac{3\pi}{4}$

d. $\frac{\pi}{8}$

23 [Foundational] • Target: The slope of a polar curve

The limaçon $r = 7 + 14\cos\theta$ has an inner loop. Find the smaller positive $\theta \in [0, 2\pi)$ where $r = 0$.

a. $\frac{2\pi}{3}$

b. $\frac{\pi}{2}$

c. $\frac{5\pi}{6}$

d. $\frac{4\pi}{3}$

24 [Foundational] • Target: Finding horizontal and vertical tangents

What is the slope of the line described by the polar equation $\theta = \pi/6$?

a. $-\frac{\sqrt{3}}{3}$

b. 0

c. $\sqrt{3}$

d. $\frac{\sqrt{3}}{3}$

25 [Foundational] • Target: Symmetry of polar curves

Find r for the rectangular point $(20,\ 21)$.

a. 41

b. 841

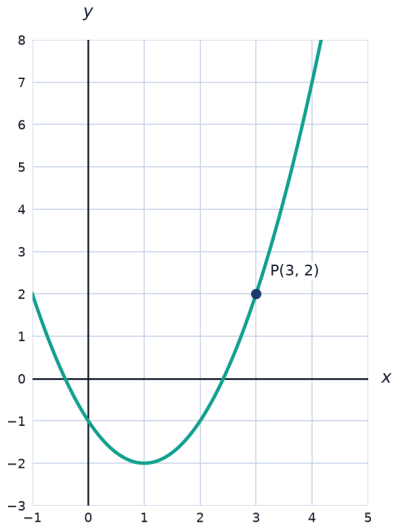
c. 29

d. 1

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26 [Foundational] • Target: More polar-to-rectangular conversions

The figure shows the graph of the function $f(x) = x^2 - 2x - 1$ and the point $P(3, 2)$ on the curve. What is the slope of the tangent line to the graph of f at point P?



a. 4

b. 2

c. 3

d. 6

27 [Foundational] • Target: More slope-of-a-polar-curve practice

Convert $r = 28\cos\theta$ to rectangular form and identify the radius of the resulting circle.

a. 56

b. 196

c. 28

d. 14

28 [Foundational] • Target: Tangent lines at the pole

Find the area of one petal of the rose $r = 3\sin(2\theta)$ (integrate θ from 0 to $\frac{\pi}{2}$).

a. $\frac{9\pi}{4}$

b. $\frac{9\pi}{16}$

c. $\frac{9\pi}{8}$

d. 9π

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Classifying polar curves by equation type

Rose curve $r = \cos(n\theta)$: number of petals when n is even:

a. n

b. $2n$

c. $n + 1$

d. $4n$

30 [Intermediate] • Target: Number of petals in rose curves

Rose curve $r = \cos(n\theta)$: number of petals when n is odd:

a. n

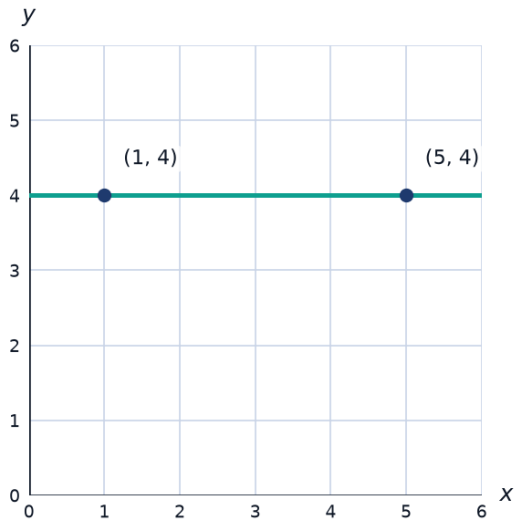
b. $2n$

c. $n/2$

d. $4n$

31 [Intermediate] • Target: Graphing polar curves point-by-point

The graph of a function $f'(x)$ is shown below. It is given that $f(x)$ is a function such that $f(1) = 5$. The area between the curve $y = f'(x)$, the x-axis, and the vertical lines $x = 1$ and $x = 5$ is represented by the shaded region. If the area of the shaded region is 16 square units, determine the value of $f(5)$.



a. 17

b. 21

c. 9

d. 11

32 [Intermediate] • Target: Concavity considerations for polar curves

Find the area of one petal of $r = 1\cos(3\theta)$ (integrate θ from $-\frac{\pi}{6}$ to $\frac{\pi}{6}$).

a. $\frac{\pi}{12}$

b. π

c. $\frac{\pi}{6}$

d. $\frac{\pi}{24}$

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33 [Intermediate] • Target: Converting between polar and rectangular coordinates

A sector of the circle $r = 10$ sweeps from $\theta = 0$ to $\theta = \pi/6$. Find its area.

a. $\frac{50\pi}{3}$

b. $\frac{25\pi}{3}$

c. $\frac{25\pi}{6}$

d. $\frac{5\pi}{3}$

34 [Intermediate] • Target: Identifying and sketching common polar curves

Find the area of the sector of $r = 3$ from $\theta = 0$ to $\theta = \pi/4$.

a. $\frac{31\pi}{24}$

b. $\frac{9\pi}{4}$

c. $\frac{9\pi}{8}$

d. $\frac{9\pi}{16}$

35 [Intermediate] • Target: The slope of a polar curve

Find the total area enclosed by the cardioid $r = 2(1 + \cos\theta)$.

a. 6π

b. 3π

c. 4π

d. 12π

36 [Intermediate] • Target: Finding horizontal and vertical tangents

Find the area inside $r = 11$ but outside $r = 6$.

a. 121π

b. 85π

c. 25π

d. $\frac{85\pi}{2}$

37 [Intermediate] • Target: Symmetry of polar curves

The slope field for a differential equation is shown above. The differential equation is of the form $\frac{dy}{dx} = ky$ for some constant k. If the solution curve passes through the point $(0, 1)$ and $(1, 2)$, find the value of k.

a. $\ln 2$

b. 2

c. $\frac{1}{2}$

d. 1

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: More polar-to-rectangular conversions • Cross-Lesson: 2.01 Euler's Method and Logistic Growth

Parametric tangent at $t = t_0$: slope $m = \frac{y'(t_0)}{x'(t_0)}$. If $x'(t_0) = y'(t_0) = 0$, the point is:

a. always a cusp

b. regular

c. a node

d. a singular/cusp point — need L'Hôpital

39 [Challenging] • Target: More slope-of-a-polar-curve practice • Cross-Lesson: 3.01 Parametric Equations

Tangent line to parametric curve is horizontal when:

a. $dx/dt = 0$

b. $dy/dt = 0$ and $dx/dte0$

c. $t = 0$

d. $dy/dt = dx/dt$

40 [Challenging] • Target: Tangent lines at the pole • Cross-Lesson: 3.02 Vector-Valued Functions

Tangent line to parametric curve is vertical when:

a. $dy/dt = 0$

b. $dx/dt = 0$ and $dy/dte0$

c. $dx/dt = dy/dt$

d. $t = 0$

41 [Challenging] • Target: Classifying polar curves by equation type • Cross-Lesson: 4.02 Area in Polar Coordinates

At which point does $f(x) = x^2 - 4x$ have a horizontal tangent?

a. $x = 0$

b. $x = 2$

c. $x = 4$

d. $x = -2$

42 [Challenging] • Target: Number of petals in rose curves • Cross-Lesson: 5.01 Sequences and Basic Series Convergence Tests

Tangent lines at origin to $r = \sin 3\theta$ (3-petal rose):

a. $\theta = 0, \pi/3, 2\pi/3$

b. $\theta = 0, \pi/6, \pi/3$

c. $y = 0$ only

d. 3 lines at $\theta = 0, \pi/3, 2\pi/3$

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43 [Challenging] • Target: Graphing polar curves point-by-point • Cross-Lesson: 5.02 Ratio, Root, and Alternating Series Tests

For the differential equation $\frac{dy}{dx} = 1x - 5y$, find the slope of the solution curve at the point $(3, 2)$.

a. -9

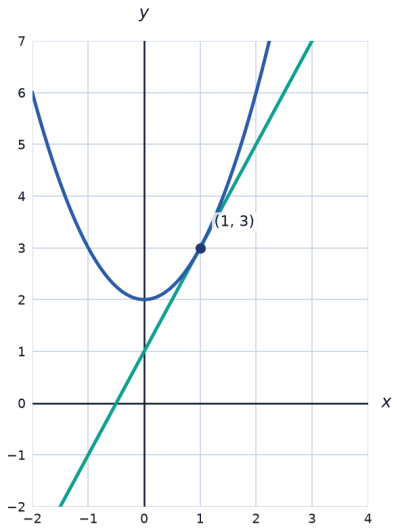
b. -5

c. 7

d. -7

44 [Challenging] • Target: Concavity considerations for polar curves • Cross-Lesson: 6.01 Power Series and Interval of Convergence

The graph of a differentiable function $f(x)$ is shown below. The function has a tangent line at $x = 1$ with equation $y = 2x + c$. If $f(1) = 3$, find the value of c.



a. 1

b. 2

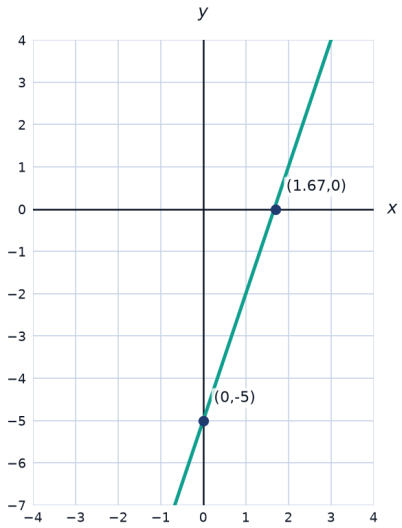
c. 3

d. 4

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45 **[Challenging]** • **Target: Converting between polar and rectangular coordinates** • **Cross-Lesson: 6.02 Taylor and Maclaurin Series**

The slope of $y = 3x - 5$ at any point is:



a. -5

b. 0

c. 3

d. undefined

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